

Haoyu Dong

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Research Profile

Ph.D. candidate in Electrical & Computer Engineering at Duke University, specializing in foundation models, LLM agents, multimodal learning, and data-centric AI. Industry research experience at Meta and Siemens Healthineers, developing algorithm-evolution methods for LLM agents and in-context learning systems for 3D medical image segmentation. Author of 33 research works in medical AI, computer vision, and multimodal learning.

Industry Experience

Meta

May 2026 – Sep. 2026

Software Engineer Intern

Menlo Park, CA

- Developed branching search for recursive self-improvement of LLM agent harnesses, evolving development objectives and proposal policies from execution feedback to discover complementary reasoning and coding strategies.
- Built a router to select among complementary harnesses, achieving relative gains over Meta-Harness of **34.8%** on Olympiad-level math, **11.6%** on Terminal-Bench 2.0, and **3.8%** on SWE-bench Lite; selected all harnesses using development data alone.

Siemens Healthineers

Jun. 2025 – Aug. 2025

AI Research Intern

Princeton, NJ

- Developed a 3D nnU-Net in-context segmentation framework with multi-stage cross-attention, reaching **52.1% Dice** on held-out unseen anatomies versus **36.7%** for the strongest prior ICL baseline, without retraining for new tasks.
- Curated **14,934 CT volumes** spanning **92 anatomical structures**; conducted the first systematic 3D support-selection study, revealing a **21-point Dice** range. Co-inventor on the resulting **patent-pending** technology assigned to Siemens Medical Solutions USA, Inc.

Gradient Health

May 2020 – Sep. 2020

Fellowship Recipient

Durham, NC

- Built a private multimodal dataset of diagnostic images and paired radiology reports from **2.6M patients**; developed a language-informed BYOL pretraining pipeline that improved accuracy by **3 percentage points** over a ResNet baseline.

Selected Research Experience

Mazurowski's Lab, Duke University

Jul. 2022 – Present

Graduate Research Associate

Durham, NC

- Pretrained **MRI-CORE** on **7M MRI slices** using modern self-supervised objectives, improving few-shot downstream segmentation by **5 Dice points** over SAM; currently developing a 3D multimodal breast MRI foundation model that aligns volumetric imaging with radiology reports.
- Led a broad evaluation of promptable segmentation across **19 datasets, 5 modalities, and 18 body locations**, establishing data-efficient interactive workflows that transfer across scanners and sites (*Medical Image Analysis*, 2023).
- Developed **InTEnt**, a single-image test-time adaptation method using integrated entropy weighting to adjust batch-normalization statistics, improving out-of-domain segmentation by **2.9 Dice points** over the strongest comparator (CVPR 2024 Workshop Oral).
- Co-developed foundation-model adaptation guidelines spanning prompt design, parameter-efficient adaptation, and light fine-tuning, and contributed to universal anatomical segmentation models across locations and acquisition protocols.

Mazurowski's Lab, Duke University

Jun. 2021 – Jun. 2022

Associate in Research

Durham, NC

- Proposed **SWSSL** for anomaly detection in high-resolution breast tomosynthesis and chest X-ray, surpassing prior state of the art by **5.0 AUROC points** on tomosynthesis (*IEEE Transactions on Medical Imaging*, 2023).

Sapiro's Lab, Duke University

Sep. 2019 – May 2021

Graduate Research Assistant

Durham, NC

- Built a visual-semantic space for text-conditioned image retrieval using iterative point alignment and manifold-distance search, improving recall by **2 points** on FashionIQ, **5 points** on CSS, and **30 points** on a new dataset (CVPR 2021 Workshop).

Selected Publications

- **Haoyu Dong**, Han Liu, Riqiang Gao, Bogdan Georgescu, Guillaume Chabin, Jianing Wang, and Sasa Grbić. “Revisiting In-Context Learning for 3D CT Segmentation of Unseen Anatomies.” *In submission*.
- **Haoyu Dong**, Yuhang Zhou, Zihao Lin, Yifan Wu, Bo Peng, Mingyi Wang, Xiangjun Fan, Lizhu Zhang, and Zhuokai Zhao. “Branching Search with Evolving Objectives and Policies for Agent Harness Optimization.” *In submission*.
- Zihao Lin, **Haoyu Dong**, Haobo Xu, Chi-Chih Chang, Xuying Ning, Yinglong Xia, Shuang Yang, Zhuokai Zhao, Xinying Song, Lifu Huang, Yang Song, and Qifan Wang. “VideoHarness-RSE: Recursive Structured Evolution of Harnesses for Video Understanding.” *In submission*.
- Hanxue Gu*, **Haoyu Dong***, Pedro R. A. S. Bassi, Wenxuan Li, Alan Yuille, Zongwei Zhou, Kang Wang, and Yang Yang. “Reports Are Not Captions: Understanding and Designing Report Supervision for Label-Efficient 3D Lesion Segmentation.” *In submission*.
- **Haoyu Dong**, Yuwen Chen, Hanxue Gu, Nicholas Konz, Yaqian Chen, Qihang Li, and Maciej A. Mazurowski. “MRI-CORE: A Foundation Model for Magnetic Resonance Imaging.” *Preprint*, 2025.
- Yuheng Li, Yuan Gao, **Haoyu Dong**, Yuxiang Lai, Shansong Wang, Mojtaba Safari, James E. Bacia, and Xiaofeng Yang. “Universal CT Representations from Anatomy to Disease Phenotype through Stage-wise Pretraining.” *npj Digital Medicine*, 2026.
- Hanxue Gu*, **Haoyu Dong***, Jichen Yang, and Maciej A. Mazurowski. “How to build the best medical image segmentation algorithm using foundation models: a comprehensive empirical study with Segment Anything Model.” *Machine Learning for Biomedical Imaging*, 2025.
- **Haoyu Dong**, Nicholas Konz, Hanxue Gu, and Maciej A. Mazurowski. “Medical image segmentation with InTEnt: Integrated entropy weighting for single image test-time adaptation.” *CVPR Workshops*, 2024.
- Maciej A. Mazurowski, **Haoyu Dong**, Hanxue Gu, Jichen Yang, Nicholas Konz, and Yixin Zhang. “Segment anything model for medical image analysis: an experimental study.” *Medical Image Analysis*, 2023.
- **Haoyu Dong**, Yifan Zhang, Hanxue Gu, Nicholas Konz, Yixin Zhang, and Maciej A. Mazurowski. “SWSSL: Sliding window-based self-supervised learning for anomaly detection in high-resolution images.” *IEEE Transactions on Medical Imaging*, 2023.

* Equal contribution. Full publication list available on Google Scholar.

Education

Duke University <i>Ph.D., Electrical & Computer Engineering</i>	Expected May 2027 <i>Durham, NC</i>
Duke University <i>M.S., Computer Science</i>	Jun. 2021 <i>Durham, NC</i>
University of California, San Diego <i>B.S., Mathematics–Computer Science; B.S., Cognitive Science</i>	Jun. 2019 <i>La Jolla, CA</i>

Technical Skills

Programming: Python, C++

Machine Learning: PyTorch, distributed training, reinforcement learning, multimodal modeling

Infrastructure: Linux, Git, Docker, Slurm, cloud platforms

Academic Service

Area Chair: GenAI4Health Workshop at NeurIPS 2026

Conference Reviewer: ICLR, NeurIPS, ICML, CVPR, ICCV, ECCV, MICCAI, MIDL

Journal Reviewer: IEEE TMI (Distinguished Reviewer, Silver Level 2023–24), IEEE JBHI, IEEE TNNLS, IEEE TBME, Medical Image Analysis, npj Digital Medicine, npj Artificial Intelligence